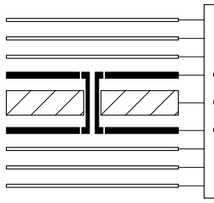
HMI Board Fabrication Document

Layer Stack Legend

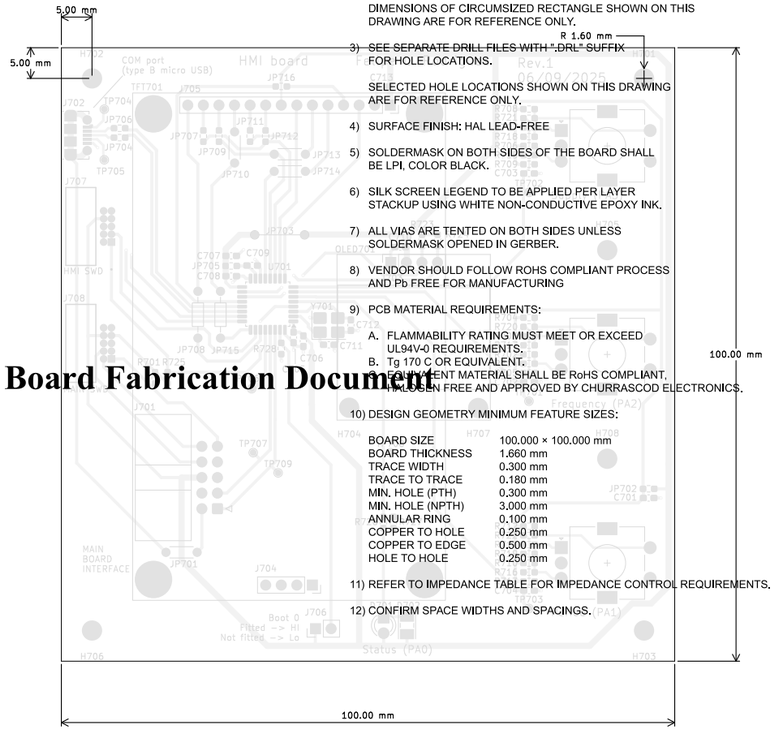


Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Core		1.48mm	FR4_7628	Dielectric	
Copper	**L6 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL LEAD-FREE

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.300 mm
TRACE TO TRACE	0.180 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	3.000 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.250 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

Top Fabrication (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b04e708
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 1 of 8

HMI Board Fabrication Document

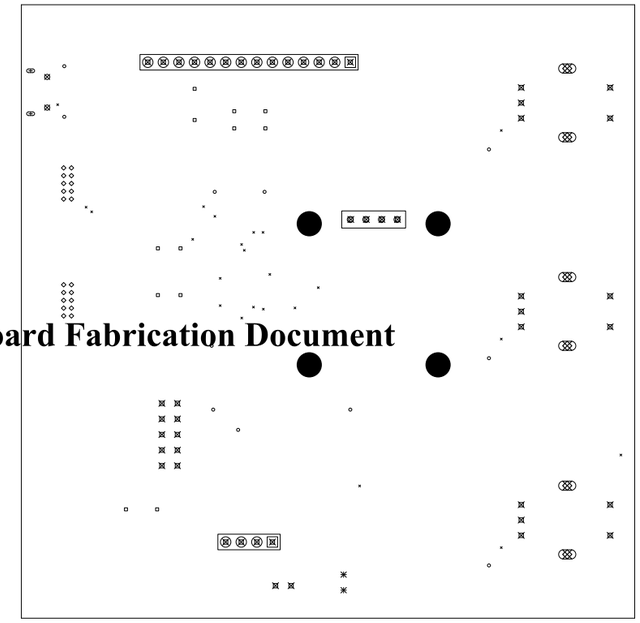
Bottom Fabrication (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b04e708
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 8

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	23	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
○	11	0.50mm (19.69mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
+	2	0.60mm (23.62mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
□	12	0.70mm (27.56mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
◇	20	0.74mm (29.13mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	2	0.85mm (33.46mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
✳	2	0.90mm (35.43mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	49	1.00mm (39.37mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	6	1.50mm (59.06mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
	Total 127					



HMI Board Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

ChurrasCod Electronics

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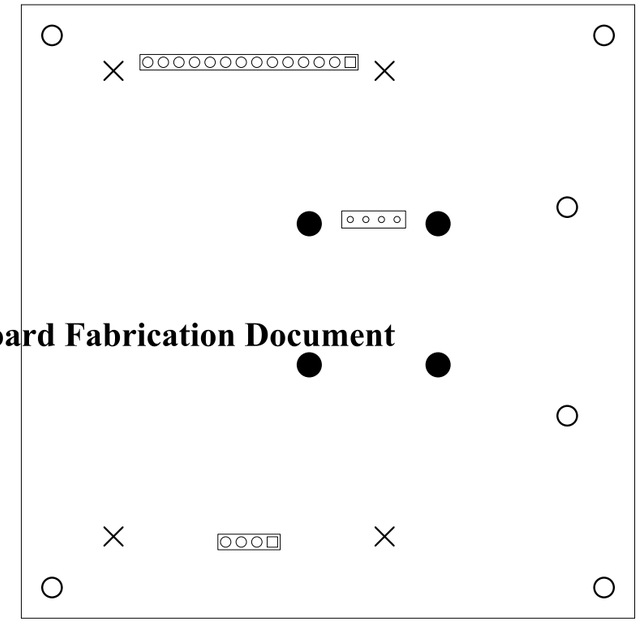
Sheet Title: Drill Drawing (L1 - L2)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 8

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	3.00mm (118.11mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
○	8	3.20mm (125.98mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
	Total 12					



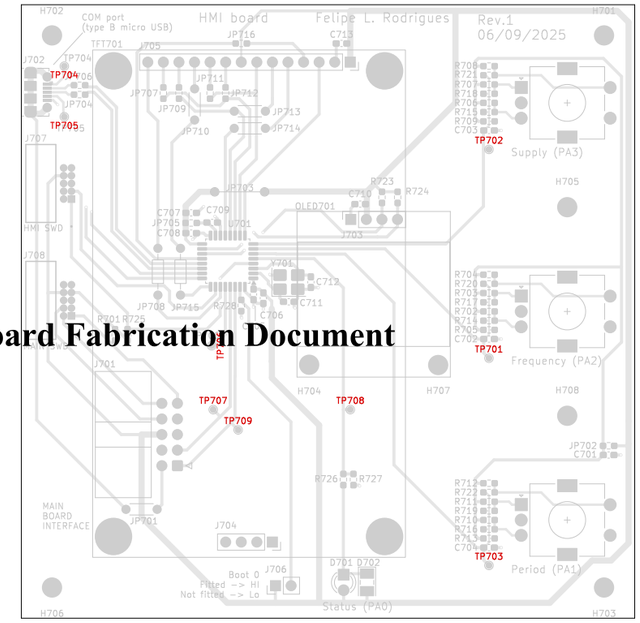
HMI Board Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b04e708
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Drill Drawing (L1 - L2)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 4 of 8

HMI Board Fabrication Document



Ref.	Net	X [mm]	Y [mm]
TP701	FREQ	86.24	52.38
TP702	SUP	86.24	86.42
TP703	PER	86.24	18.60
TP704	Net-(J702-VBUS)	17.00	99.98
TP705	Net-(J702-ID)	17.00	91.73
TP706	Net-(U701-PB5)	41.03	54.41
TP707	LV_TX	41.28	44.00
TP708	ST_LED	63.63	44.00
TP709	LV_RX	45.34	40.70

Ref.	Net	X [mm]	Y [mm]

Top Test Points (Scale 1:1)

Y
X

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Sheet Path:		Reviewer:	Size: A3	Sheet: 5 of 8

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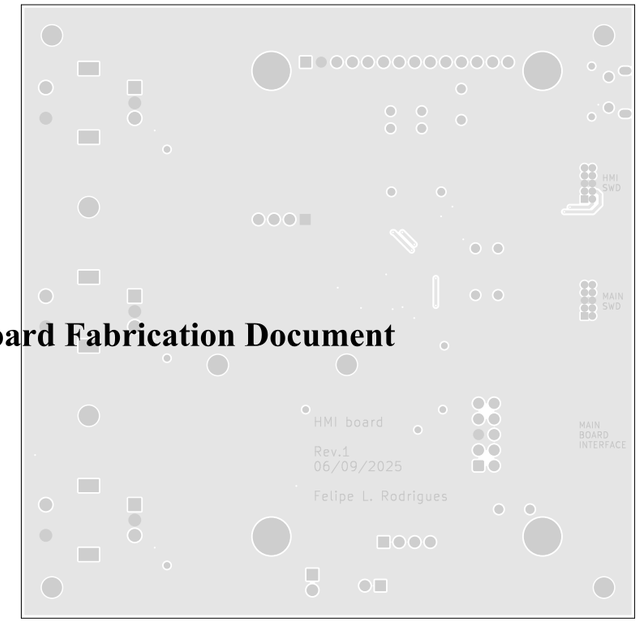

Ref.	Net	X [mm]	Y [mm]

Bottom Test Points (Scale 1:1)

Y
X

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b04e708
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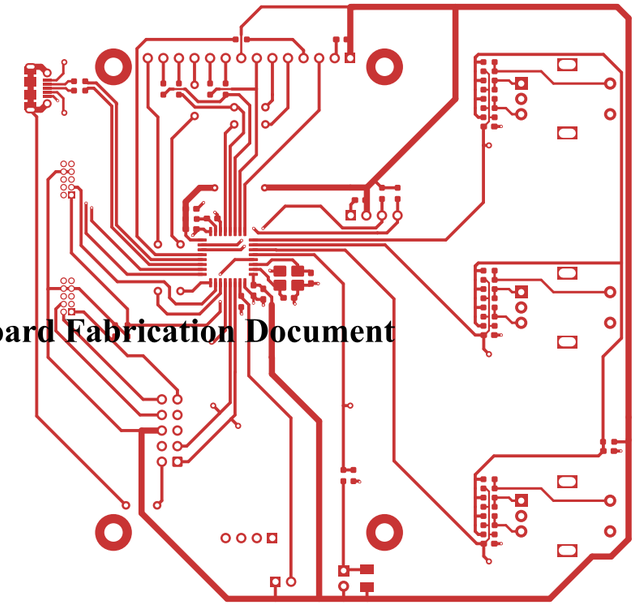
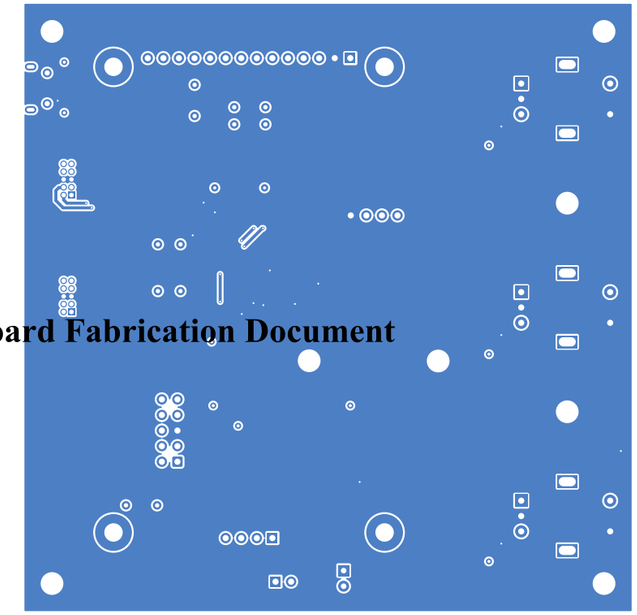
HMI Board Fabrication Document



L1 (Sig, PWR) (Scale 1:1)

ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics	Variant: PRELIMINARY	Git Hash: b04e708
		Board Name: HMI Board	Project Name: 1 kV Strobe	
	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 7 of 8

HMI Board Fabrication Document



L6 (Sig, PWR) (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b04e708
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: L6 (Sig, PWR) (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 8 of 8